Exact Inference in Bayesian Networks

Block 2 Day 2

Semester 1, 2026–2027 — Teaching Week 4

Session Overview

Session objective: Strengthen exact inference through worked enumeration, elimination-order reasoning, and complexity awareness without copying likely exam arithmetic exactly.
Timing target (min): 71
Topic anchors: exact inference; enumeration; variable elimination; elimination order; treewidth
Padlet board: https://padlet.com/qmul/b2-d2-cxmnkit5wkn83isk

Exercises

Q1 Core Concept Check

Question type Concept → **Working mode** Pair → **Expected time** 9 min

Prompt

Contrast exact and approximate inference using three ideas: guarantee of the answer, computational cost, and when each approach is preferable in a live system.

Task details

- **Type:** concept
- **Mode:** pair
- **Time:** 9 min
- **Deliverable:** comparison table
- **Assessment focus:** contrast exact and approximate inference in practical terms
- **Expected length:** 3 rows

How to submit

Post one pair Padlet comment as a three-row comparison table or numbered list.

Discussion in class

Why do teams still use exact inference if it can become expensive?

Why this helps

Useful for short comparison questions before the mathematical work begins.

References

- Russell & Norvig (2021) AIMA 4e, Ch.13, pp.412-460
- Poole & Mackworth (2023) 3e, Ch.9, pp.375-458

Q2 Structured Problem

Question type	Working mode	Expected time
Problem	Individual	13 min

Prompt

Use the Alarm network numbers from class: P(B)=0.001, P(E)=0.002, P(A|B,E)=0.95, P(A|B,~E)=0.94, P(A|~B,E)=0.29, P(A|$_{\text{B,}}$E)=0.001, P(J|A)=0.90, P(J|~A)=0.05, P(M|A)=0.70, P(M|~A)=0.01. Compute P(E=true | J=true, M=false) by enumeration, including normalization.

Formula reminder

$$P(X \mid e) = \alpha\, P(X, e)$$

$$P(X, e) = \sum_{h} P(X, e, h)$$

Use this when you must compute two unnormalised scores and then normalise.

Task details

- **Type:** problem
- **Mode:** individual
- **Time:** 13 min
- **Deliverable:** worked enumeration
- **Assessment focus:** carry out one exact inference query by enumeration with normalization
- **Expected length:** 5 calculation lines

How to submit

Post your own Padlet comment showing the two unnormalized scores, the normalization step, and the final probability.

Discussion in class

Which hidden variables do you need to sum out, and why?

Why this helps

Direct practice for exact enumeration while using a different target query from the exam set.

References

- Russell & Norvig (2021) AIMA 4e, Ch.13, pp.412-460
- Poole & Mackworth (2023) 3e, Ch.9, pp.375-458

Q3 Structured Problem

Question type	Working mode	Expected time
Problem	Individual	12 min

Prompt

Consider a network with edges Season -> Pollen, Season -> Flu, Pollen -> Sneezing, Flu -> Fever, Flu -> Sneezing. For the query P(Flu | Sneezing=true), choose an elimination order for Season, Pollen, and Fever, and explain briefly what factor would become awkward if you eliminated the wrong variable too early.

Task details

- **Type:** problem
- **Mode:** individual
- **Time:** 12 min
- **Deliverable:** elimination trace
- **Assessment focus:** reason about factor growth under variable elimination
- **Expected length:** ordered list + 3 short notes

How to submit

Post your own Padlet comment with the order and three short notes on factor growth.

Discussion in class

Which hidden variable acts like the most dangerous hub in this network?

Why this helps

Useful for variable-elimination reasoning without copying the exact financial-risk exam scenario.

References

- Russell & Norvig (2021) AIMA 4e, Ch.13, pp.412-460
- Poole & Mackworth (2023) 3e, Ch.9, pp.375-458

Q4 Collaborative Mini-Case

Question type Case → **Working mode** Group → **Expected time** 14 min

Prompt

You are comparing two Bayesian networks for a live monitoring system. Network A is sparse and tree-like. Network B has many cross-links between risk variables. Write a short memo explaining which one is safer for exact inference under latency pressure, and why.

Task details

- **Type:** case
- **Mode:** group
- **Time:** 14 min
- **Deliverable:** complexity memo
- **Assessment focus:** explain why one network is easier than another for exact inference
- **Expected length:** 4 bullets

How to submit

Post one group Padlet comment with four bullets and one recommended operational choice.

Discussion in class

Is graph size or graph shape the more dangerous issue for exact inference?

Why this helps

Supports exam-style complexity reasoning without turning the prompt into a definition-only question.

References

- Russell & Norvig (2021) AIMA 4e, Ch.13, pp.412-460
- Poole & Mackworth (2023) 3e, Ch.9, pp.375-458

Q5 Exam-Style Constructed Response

Question type Exam → **Working mode** Pair → **Expected time** 13 min

Prompt

Explain treewidth in simple language to an engineering manager who only cares about runtime. Then explain one practical thing the team can do when exact inference becomes too costly.

Task details

- **Type:** exam
- **Mode:** pair
- **Time:** 13 min
- **Deliverable:** written analysis
- **Assessment focus:** explain treewidth and one mitigation in plain language
- **Expected length:** 180-230 words

How to submit

Post one pair Padlet comment with a plain-language explanation, one operational consequence, and one mitigation.

Discussion in class

Is switching to approximate inference the only sensible mitigation?

Why this helps

Matches exam-style complexity explanation, but the audience and framing are different from the paper question.

References

- Russell & Norvig (2021) AIMA 4e, Ch.13, pp.412-460
- Poole & Mackworth (2023) 3e, Ch.9, pp.375-458

Q6 Stretch Extension

Question type	Working mode	Expected time
Stretch	Group	10 min

Prompt

A fraud-monitoring system must respond within 300 milliseconds. Write a short fallback policy saying when the system should stop using exact inference, what it should use instead, and how the team should monitor that fallback.

Task details

- **Type:** stretch
- **Mode:** group
- **Time:** 10 min
- **Deliverable:** fallback policy
- **Assessment focus:** define an operational switch from exact to approximate inference
- **Expected length:** 4 policy rules

How to submit

Post one group Padlet comment with a threshold, a fallback method class, and one monitoring rule.

Discussion in class

What signal should trigger the switch first: latency, memory use, or factor size warning?

Why this helps

Good extension for deployment questions that connect inference choice to live operational constraints.

References

- Russell & Norvig (2021) AIMA 4e, Ch.13, pp.412-460
- Poole & Mackworth (2023) 3e, Ch.9, pp.375-458

Submission Instructions

Use the seeded Padlet posts for Q1–Q6. Q2 and Q3 are individual. Q1 and Q5 are pair tasks. Q4 and Q6 should be one joint group comment. Start each comment with your names or group label.

Whole-Class Debrief Prompts

- Which step of enumeration created the most repeated work?
- Why is elimination order a design decision rather than a purely mechanical one?
- When should a real system stop insisting on exact inference?

Exam Transfer Note (Day-Level)

Maps to exam questions on enumeration, elimination order, factor growth, treewidth, and exact-inference trade-offs.